DIGITAL SAFETY TOOLKIT FOR SCHOOLS (CYBER-HYGIENE KIT)

A Capstone Project Collaboration:

Cybersecurity & Technical Project Management

TECH4DEV

THE DIGITAL GUARDIAN - GROUP 42

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1. PROJECT OVERVIEW

The Digital Safety Toolkit for Schools (Cyber-hygiene) is an educational and cybersecurity awareness project designed to equip students, teachers and school administrators with essential knowledge and tools for safe digital practices in the modern learning environment.

The project was developed in direct response to a documented and growing crisis in educational digital safety measures. Schools across Africa and beyond are rapidly digitizing their classrooms, yet three critical vulnerabilities continue to expose students and educators to preventable harm:

- Students rely on weak, reused passwords making every account a potential entry point for attackers.
- Invisible trackers, advertisements and malicious scripts harvest student data without consent as they browse educational platforms.
- Teachers, though willing, lack the practical tools and knowledge to create a digitally secure classroom environment.

Together, these vulnerabilities mean that every time a student goes online to learn, they face real and present digital threats including phishing attacks, identity theft, cyberbullying and data exploitation. This directly undermines the promise of quality, inclusive, and safe digital education.

The Toolkit addresses this problem by delivering three integrated pillars of cyber-hygiene: a Password Manager to eliminate credential risk, Browser Safety Extensions to neutralise environmental threats, and Privacy Checklists and Training Guides to build lasting behavioural awareness in both staff and students. The solution was developed collaboratively by the Cybersecurity Team (who designed and validated the technical components in a Kali Linux lab environment) and the Technical Project Management Team (who structured the project, managed delivery and produced all user-facing documentation).

Objectives:

1. Improve digital safety awareness in schools (ages 12-18), teachers and school administrators
2. Provide practical and easy-to-deploy cybersecurity tools suitable for school environments.
3. Build responsible, long-term online behaviour through guided training and privacy

checklists.

4. Reduce exposure to phishing, identity theft, data harvesting and account compromise.
5. Align digital safety education with UN SDG 4: Quality Education.

2. TARGET USERS

The toolkit was designed to serve three distinct user groups within the school ecosystem, each with tailored content ensuring relevance, ease of adoption and measurable impact:

User Group	Role in the Toolkit	Key Needs Addressed
Students (Ages 12-18)	Primary users: direct learners and beneficiaries of all three toolkit pillars.	Safe internet habits, password security, protection from hackers and phishing, responsible online behaviour.
Teachers	Facilitators: trained to deliver digital safety education and enforce safe practices in the classroom.	Practical, non-technical guides; ability to audit classroom apps; confidence to lead digital safety discussions.
School Administrators	Policy implementers: responsible for deploying toolkit tools school-wide and monitoring compliance.	Clear implementation framework, measurable outcomes, tools that integrate into existing school systems.

3. SOLUTION EXPLANATION

The Cyber-Hygiene Kit is a proactive defense system designed to sanitize the digital environment of a school. Much like physical hygiene prevents the spread of disease, this kit focuses on "cleaning" the digital habits and tools of students and teachers to prevent data "infections" and identity theft. The framework shifts responsibility from the individual's unreliable memory to automated and high-standard security protocols.

3.1 Identity Hygiene: The Password Manager

Think of this as a Sterile Vault for credentials. In a typical unprotected environment, a student uses one weak password across multiple platforms. If one site is breached, the infection spreads to every other account, a phenomenon known as Credential Decay.

- **The Mechanism:** Generates high-entropy, unique "digital DNA" for every single login automatically.
- **Impact:** Eliminates Credential Decay and ensures that even if a third-party site is hacked, the student's academic records and identity remain completely untouched.

3.2 Environmental Hygiene: Browser Safety Extensions

These act as a Digital Air Filtration System. The internet is filled with "Digital Dust" trackers, malicious scripts and invasive advertisements that harvest student data without their knowledge or consent.

- **The Mechanism:** Lightweight browser add-ons (scrubbers) sit between the user and the web, automatically neutralising trackers and blocking malicious code before it loads.
- **Impact:** Creates a Decontaminated Learning Zone, students can research and learn freely in a private, distraction-free environment, without their browsing habits being monitored by external entities.

3.3 Behavioral Hygiene: Privacy Checklists & Training

Hygiene is a habit, not just a tool. Even the best locks are useless if the windows are left open. These guides provide the Standard Operating Procedures (SOPs) for maintaining a secure classroom.

- **The Mechanism**: They translate complex technical settings into plain-language "safety checks." Teachers can independently audit school apps and social media permissions to "harden" their classroom's digital perimeter.
- **Impact**: Builds Long-Term Cyber-Awareness by converting both staff and students from passive targets into active, informed defenders of their own digital safety.

The Three Pillars work as an integrated system. The Password Manager protects identity. The Browser Extensions protect the environment. The Training Guides protect behaviour. Together, they create a self-reinforcing cycle of digital safety that becomes embedded in the culture of the school.

4. TECHNOLOGY USED

The project utilised a carefully selected combination of professional tools to ensure security, collaboration and high-quality output across all phases of development:

Category	Tool / Platform	Purpose
Project Management	ClickUp	Task tracking, workflow coordination and milestone monitoring across both teams.
Security Lab / Testing	Kali Linux	Controlled environment for validating toolkit tools, running simulated threat tests on student apps and browser extensions.
Documentation	Microsoft Word and Google Docs	Proposal writing, guide development, documentation and version control across the project.
Presentation	PowerPoint and Google Slides	Demo Day slide deck creation and stakeholder presentation preparation.
Storage & Collaboration	Google Drive and GitHub	Cloud-based file sharing, team collaboration, version control and artefact storage.

5. AI COMPONENT EXPLANATION

Artificial Intelligence was embedded throughout the entire development lifecycle of this project, not as a shortcut, but as a strategic partner at every stage, from problem discovery through to final presentation. The team deployed four distinct AI tools, each assigned a specific and deliberate role:

5.1 ChatGPT — Drafting and Content Design

- **Tool Used:** Acted as the core drafting assistant for the Project Scope and Project Charter.
- **How Used:** Generated clear, student-friendly explanations of complex cybersecurity concepts for all toolkit materials, ensuring that password security, phishing risks and browser safety were communicated in language accessible to students (ages 12-18), teachers, and non-technical administrators.
- **Part of Solution Supported:** Defined clear project boundaries, ensuring deliverables were scoped appropriately. Produced the student-friendly guide content and checklist language embedded in Pillar 3 of the toolkit.

5.2 Claude AI — Research and Strategic Synthesis

- **Tool Used:** Conducted deep research and strategic analysis to bridge technical security findings with clear project documentation.
- **How Used:** Used to synthesise cybersecurity research literature, analyse lab findings from the Kali Linux environment, transform raw technical data into professionally structured proposals and documentation narratives.
- **Part of Solution Supported:** Ensured the solution narrative aligned cybersecurity team findings with stakeholder communication needs and refined the connection between the Three Pillars framework and SDG 4 goals.

5.3 Gemini AI — Problem Discovery and SDG Alignment

- **Tool Used:** Served as the primary research engine for problem identification and SDG alignment.
- **How Used:** Investigated current cybersecurity threat trends within Nigerian and global educational sectors and mapped the project's solution directly to the specific targets within SDG 4 (Quality Education).
- **Part of Solution Supported:** Instrumental in identifying the three core vulnerabilities (weak passwords, data harvesting, unprepared educators) and structuring the solution into the Three Pillars of Cyber-Hygiene framework, ensuring the project was both technically sound and socially impactful.

5.4 Gamma AI — Visual Presentation Design

- **Tool Used:** Dedicated AI-powered presentation design platform.
- **How Used:** Used to transform the technical framework, research data and written documentation into a professional, visually compelling slide deck for the Demo Day presentation.
- **Part of Solution Supported:** Supported the final stakeholder communication layer of the project, ensuring that complex cybersecurity concepts were translated into clear, engaging visuals accessible to all audiences.

6. DEVELOPMENT PROCESS

The project followed an Agile-based iterative development approach, structured into three core phases. This methodology enabled both teams to collaborate efficiently, adapt to findings and deliver a polished, validated toolkit within the programme timeline.

PHASES:

Initiation and Strategic Alignment

- **Defined project goals:** Established the core objective of sanitising school digital habits and building a secure online learning environment.
- **Identified stakeholders:** Mapped the specific needs of students (ages 12–18), teachers, and school administrators through structured user analysis.
- **Conducted research on digital safety:** Investigated current cybersecurity trends, with focus on phishing, identity theft and data harvesting in educational sectors, supported by Gemini AI and Claude AI.
- **Aligned with UN SDG 4:** Anchored the project in Quality Education by demonstrating that cyber hygiene is a prerequisite for safe, inclusive digital learning.

Planning & Laboratory Setup

The initial phase focused on establishing a solid foundation for the project by defining its scope and preparing the technical environment required for validation.

The Technical Project Management team developed the Project Charter, defining objectives, deliverables and success criteria. ClickUp was configured for task tracking and sprint management across both teams, ensuring accountability and visibility throughout the project.

Simultaneously, the Cybersecurity Team established a controlled testing environment in Kali Linux. Key stress test targets were identified including commonly used student applications and educational websites to ensure all security validations reflected real-world conditions.

The Three Pillars framework was finalised at this stage, with each pillar assigned to specific team members for development and validation.

Phase 3: Design, Build & Validation

Technical toolkit components were validated through simulated threat testing in the Kali Linux lab. Results were documented through logs and screenshots for credibility. Findings were then translated into user-facing materials using AI tools; producing guides, privacy checklists, posters in plain and accessible language.

The TPM team simultaneously created structured content templates, managed documentation and coordinated a comprehensive risk evaluation. The project concluded with a full review by both teams to ensure deliverables met stakeholder requirements before packaging for Demo Day.

7. LESSONS LEARNED

The following key insights were gained during the development and validation of the Digital Safety Toolkit:

Lesson	Detail & Implication
User-Centred Design is Critical	All materials must be simple, visual and relatable for students. Technical language consistently reduced engagement; plain-language equivalents improved comprehension and adoption rates significantly.
Early Stakeholder Engagement is Essential	Involving teachers in the design phase rather than presenting a completed solution led to more practical, classroom-ready materials and higher likelihood of real-world adoption.
Tools Alone Are Never Enough	Deploying browser extensions and password managers without accompanying training left users uncertain and underconfident. The training guides proved to be the glue that made the technical tools effective.

Simplicity Drives Adoption	Overly technical checklists and guides were consistently avoided or abandoned. The most effective materials were those that could be completed in under 10 minutes with no prior IT knowledge.
AI Accelerates, But Humans Must Validate	AI tools dramatically sped up content creation, research synthesis, and presentation design. However, all AI-generated content required human review to ensure accuracy, contextual relevance and appropriateness for the target audience.
Pilot Testing Reveals the Unexpected	Lab testing and simulated scenarios uncovered issues that stakeholder interviews missed. Iterative feedback loops during pilot testing significantly elevated the quality and reliability of all three toolkit pillars.

8. CONCLUSION

The Digital Safety Toolkit for Schools successfully delivers a practical, scalable and user-friendly solution to one of the most urgent and underaddressed challenges in digital education: the absence of accessible cyber-hygiene tools and awareness in schools.

By combining the technical rigour of the Cybersecurity Team's lab-validated tool selection with the structured delivery framework of the TPM Team and amplifying both with AI-assisted research, content design, and presentation tools, the project demonstrates that meaningful digital safety education can be implemented in any school without specialist IT resources.

The Three Pillars of Cyber-Hygiene: Identity Hygiene, Environmental Hygiene and Behavioural Hygiene provide a self-reinforcing framework that protects students at every layer of their digital lives: their credentials, their browsing environment and their everyday habits.